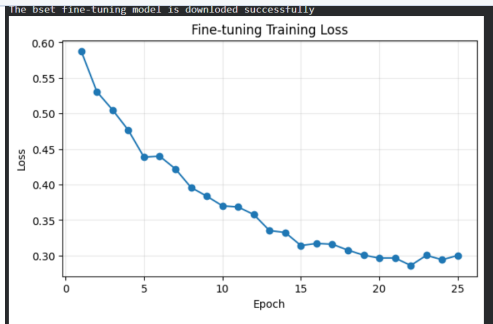


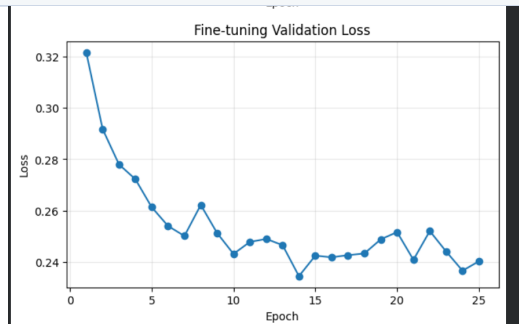
Model Benchmark & Comparison																																					
TOTAL MODELS		HIGHEST F1 SCORE		HIGHEST QWK SCORE		TOP PERFORMING MODEL																															
8		0.6919		0.8841		GeM Pooling with V2 Augemnt																															
Architectures Evaluated		Validation Dataset		Quadratic Weighted Kappa		Highest Overall QWK																															
Model Name	Architecture	Description	F1 Score	QWK	Train Loss	Validation Loss	Epochs																														
EfficentNetB3 base line	With Freezing	EffecintNetB3	0.4610			0.6500	10																														
base model + V1 augument	With Freezing	Used Focal Loss	0.5730		0.4400	0.3000	10																														
GeM Pooling with V2 Augemnt	With Freezing	Note that is was less PreProcessing	0.6125	0.8495		0.3620	40																														
GeM Pooling with V2 Augemnt	No Freeze	Image size=300, 70% for training	0.6919	0.8841	0.3000	0.2404	40																														
GeM Pooling Higher size	No Freeze Better General	Image size=384, 80% for training, 5e-4 WD	0.6488	0.8774	0.2500	0.2900	40																														
ConvNet With V1 Augmentation	With Freezing		0.6580	0.8733		0.2830	40																														
Corn V1	With Freezing	V1 augument	0.5988	0.8430		0.2989	20																														
Corn V2	With Freezing	V2 augemnt with Gem layer	0.5735	0.8532		0.2841	20																														
Corn Model Final	No Freeze Better General	Image size=384, 80% for training, 5e-4 WD	0.6338	0.8831	0.2255	0.2863	40																														
Model Performance Benchmark: F1 Score vs. QWK <table><thead><tr><th>Model</th><th>F1 Score</th><th>QWK</th></tr></thead><tbody><tr><td>EfficentNetB3 base line</td><td>0.4610</td><td></td></tr><tr><td>base model + V1 augument</td><td>0.5730</td><td></td></tr><tr><td>GeM Pooling with V2 Augemnt</td><td>0.6125</td><td>0.8495</td></tr><tr><td>GeM Pooling with V2 Augemnt</td><td>0.6919</td><td>0.8841</td></tr><tr><td>GeM Pooling Higher size</td><td>0.6488</td><td>0.8774</td></tr><tr><td>ConvNet With V1 Augmentation</td><td>0.6580</td><td>0.8733</td></tr><tr><td>Corn V1</td><td>0.5988</td><td>0.8430</td></tr><tr><td>Corn V2</td><td>0.5735</td><td>0.8532</td></tr><tr><td>Corn Model Final</td><td>0.6338</td><td>0.8831</td></tr></tbody></table>								Model	F1 Score	QWK	EfficentNetB3 base line	0.4610		base model + V1 augument	0.5730		GeM Pooling with V2 Augemnt	0.6125	0.8495	GeM Pooling with V2 Augemnt	0.6919	0.8841	GeM Pooling Higher size	0.6488	0.8774	ConvNet With V1 Augmentation	0.6580	0.8733	Corn V1	0.5988	0.8430	Corn V2	0.5735	0.8532	Corn Model Final	0.6338	0.8831
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Model Loss Curves & Training Diagnostics																																					
Training and validation loss curves for comparative analysis																																					

MODEL 1: GeM Pooling with V2 Augemnt (Candidate)

Training Loss Curve

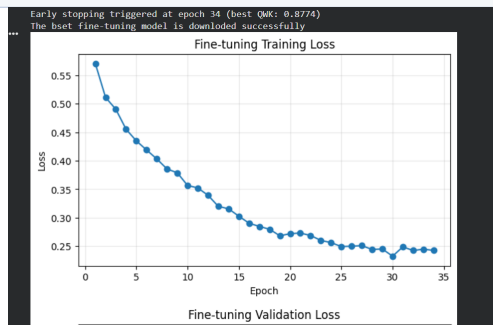


Validation Loss Curve



MODEL 2: GeM Pooling Higher size (Champion)

Training Loss Curve



Validation Loss Curve

